

The $d = 4$ Hypercube Inequality on Supports of Size at Most Five

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Abstract

We prove the $d = 4$ Hypercube Inequality for every nonnegative assignment to the 16 cube vertices whose support has size at most five. Equality within this restricted domain holds exactly when the support is contained in a cube facet. For a five-point support, the restricted degree-eight gap is reduced under the order-384 cube symmetry group. Exact enumeration gives 27 orbits, 10 affinely dependent and 17 independent; for every independent orbit representative, direct integer expansion gives a nonzero polynomial with nonnegative coefficients. A standard-library verifier reconstructs all $\binom{16}{5} = 4368$ supports, determinants, orbits, and coefficient maps. This is a sparse-boundary result only: supports of size at least six are not covered, and the original 16-variable $d = 4$ inequality remains unresolved.

1 Statement and context

Let

$$V_4 = \{0, 1\}^4, \quad F_{r,b} = \{v \in V_4 : v_r = b\} \quad (1 \leq r \leq 4, b \in \{0, 1\}).$$

For a five-element set $S = \{v_0, \dots, v_4\} \subset V_4$, use the normalized simplex volume

$$\text{Vol}(S) = 4! \text{ vol}_4(\text{conv } S) = |\det(v_1 - v_0, \dots, v_4 - v_0)|.$$

Thus $\text{Vol}(S)$ is a nonnegative integer and vanishes precisely when S is affinely dependent. For $x = (x_v)_{v \in V_4} \in \mathbb{R}_{\geq 0}^{16}$, set

$$\begin{aligned} P_4(x) &= \sum_{S \in \binom{V_4}{5}} \text{Vol}(S) \prod_{v \in S} x_v, \\ \sigma(x) &= \sum_{v \in V_4} x_v, \\ Q_4(x) &= \prod_{r=1}^4 \prod_{b=0}^1 \left(\sum_{v \in F_{r,b}} x_v \right). \end{aligned}$$

The $d = 4$ Hypercube Inequality is the universal assertion

$$P_4(x) \sigma(x)^3 \leq Q_4(x) \quad (x \in \mathbb{R}_{\geq 0}^{16}). \quad (\text{HC4})$$

Averkov, von Dichter, and Soprunov introduced this formulation, proved the $d = 3$ case, and in the consulted version stated that $d \geq 4$ remained open [1]. Their parameter convention is $d = n - 1$, so (HC4) corresponds to their geometric dimension $n = 5$, not to the proved $n = 4$ case. They also noted that the unrestricted $d = 4$ expansion has roughly 400,000 terms. The

contribution here is instead a complete exact classification of the first nontrivial sparse-support layer.

The certificate architecture has genuine methodological precedents but does not borrow a mathematical premise from them. Exact finite-library results in discrete geometry have likewise combined symmetry reduction, integer predicates, and explicit separation from the ambient open problem [2]. Exact local positivity work has similarly retained checkable certificates while limiting its conclusions to stated subdomains [3]. Here the final positivity certificates are simpler: they are coefficientwise nonnegative integer expansions. A bounded comparison found no publication of the particular five-support classification recorded below as of September 9, 2026; this is not a priority claim.

Theorem 1. *If $x \in \mathbb{R}_{\geq 0}^{16}$ and $|\text{supp } x| \leq 5$, then*

$$P_4(x)\sigma(x)^3 \leq Q_4(x).$$

Within this restricted domain, equality holds if and only if $\text{supp } x$ is contained in one of the eight facets $F_{r,b}$.

Theorem 1 makes no assertion for $|\text{supp } x| \geq 6$ and therefore does not resolve (HC4).

2 Exact reduction and proof

If $|\text{supp } x| \leq 4$, every squarefree degree-five monomial in P_4 vanishes. Now suppose $\text{supp } x = T = \{v_0, \dots, v_4\}$ and write $y_j = x_{v_j} > 0$. Only the summand indexed by T can survive in P_4 , so the restricted gap is

$$D_T(y) = \prod_{r=1}^4 \left(\sum_{j:(v_j)_r=0} y_j \right) \left(\sum_{j:(v_j)_r=1} y_j \right) - \text{Vol}(T)y_0y_1y_2y_3y_4(y_0 + \dots + y_4)^3. \quad (1)$$

If T is affinely dependent, then $\text{Vol}(T) = 0$, and (1) is a product of eight linear forms with nonnegative coefficients.

For the remaining sets, let

$$G = \{v \mapsto (v_{\pi(1)}, \dots, v_{\pi(4)}) \oplus e : \pi \in S_4, e \in \{0, 1\}^4\}.$$

This group has order $2^4 4! = 384$. It permutes the cube facets, and its linear parts are signed permutation matrices. Hence it preserves both Q_4 and the absolute determinant defining Vol , up to relabelling the five variables. It is enough to expand (1) on one representative of each G -orbit.

Table 1 is the complete enumeration. “Terms” counts nonzero monomials in D_T ; “range” gives the minimum and maximum among its nonzero coefficients. A dash marks a dependent representative. Every displayed quantity is obtained with integer arithmetic.

Table 1: The 27 symmetry orbits of five-subsets of V_4 .

No.	Representative	Orbit	Vol	Terms	Range
1	0000,0001,0010,0011,0100	192	0	–	–
2	0000,0001,0010,0011,1100	96	0	–	–
3	0000,0001,0010,0100,0111	64	0	–	–
4	0000,0001,0010,0100,1000	16	1	53	1–9
5	0000,0001,0010,0100,1001	192	1	79	1–13
6	0000,0001,0010,0100,1011	192	1	118	1–16
7	0000,0001,0010,0100,1111	64	1	165	1–18

No.	Representative	Orbit	Vol	Terms	Range
8	0000,0001,0010,0101,0110	192	0	–	–
9	0000,0001,0010,0101,1010	192	1	108	1–16
10	0000,0001,0010,0101,1011	384	1	120	1–16
11	0000,0001,0010,0101,1110	384	1	162	1–20
12	0000,0001,0010,0111,1011	96	1	125	1–17
13	0000,0001,0010,0111,1100	192	1	174	1–18
14	0000,0001,0010,0111,1101	384	1	168	1–20
15	0000,0001,0010,0111,1111	192	1	157	1–23
16	0000,0001,0010,1100,1101	192	0	–	–
17	0000,0001,0010,1100,1111	96	0	–	–
18	0000,0001,0010,1101,1110	96	0	–	–
19	0000,0001,0010,1101,1111	192	0	–	–
20	0000,0001,0110,0111,1010	192	0	–	–
21	0000,0001,0110,1010,1100	64	2	173	1–12
22	0000,0001,0110,1010,1101	192	2	237	1–18
23	0000,0001,0110,1010,1111	192	1	235	1–24
24	0000,0001,0110,1011,1110	192	1	225	1–27
25	0000,0011,0101,0110,1001	64	2	180	1–12
26	0000,0011,0101,1001,1110	16	3	246	1–12
27	0000,0011,0101,1010,1100	48	0	–	–

The orbit sizes in the table sum to $4368 = \binom{16}{5}$. There are 10 dependent orbits, containing 1360 supports, and 17 independent orbits, containing 3008 supports. The full normalized-volume census is

$$\#\{T : \text{Vol}(T) = 0, 1, 2, 3\} = (1360, 2672, 320, 16).$$

For every independent representative, exact expansion yields

$$D_T(y) = \sum_{\substack{\alpha \in \mathbb{Z}_{\geq 0}^5 \\ |\alpha|=8}} c_{T,\alpha} y^\alpha, \quad c_{T,\alpha} \in \mathbb{Z}_{\geq 0}. \quad (2)$$

and at least one coefficient is positive. The complete exponent–coefficient maps, rather than only the summaries in Table 1, are retained in the accompanying certificate.

Proof of Theorem 1. The preceding reduction proves nonnegativity when the support has at most four points or is a dependent five-set. For an independent five-set, transport it to the representative of its orbit. Equation (2) gives $D_T(y) \geq 0$; since all five y_j are positive and (2) is not the zero polynomial, in fact $D_T(y) > 0$.

It remains to classify equality. In the cases with $P_4(x) = 0$, nonnegativity of the variables gives

$$Q_4(x) = 0 \iff \sum_{v \in F_{r,b}} x_v = 0 \text{ for some } (r, b) \iff \text{supp } x \subseteq F_{r,1-b}.$$

Thus equality then holds exactly when the support lies in a facet. Independent five-point supports are strict by (2). Conversely, if the support lies in a facet, the opposite facet factor in Q_4 vanishes and all five-point determinants supported there vanish, so both sides are zero. The empty support of the zero vector is included. $\square$

3 Reproducibility and limitations

From the project root, the exact check is

```
python3 evidence/verify_five_supports.py \
--check evidence/five_support_certificate.json
```

The verifier uses the Python standard library only. It reconstructs V_4 , all 4368 five-subsets, and all 384 coordinate-permutation/complement maps. It chooses the lexicographically least

orbit image, evaluates the augmented integer determinant by the Leibniz formula, expands the eight facet factors in exponent dictionaries, subtracts the exact multinomial expansion of the second term in (1), and compares every resulting map with the serialized certificate. It also checks orbit coverage, orbit-stabilizer identities, volume invariance, and all coefficient signs. No floating-point comparison, random sampling, or numerical volume enters the certificate.

A separate fresh computational audit regenerated the certificate and also expanded all 4368 support gaps directly, using fraction-free determinant elimination rather than the certificate generator’s determinant routine. It reproduced the orbit counts, volume census, and coefficient signs. This is an additional exact check, not a claim of peer review or a formal proof-assistant verification.

The finite argument provides no sparsification, polarization, or extremal support theorem that transfers the result to six or more positive variables. Consequently, the unrestricted non-negativity of $Q_4 - P_4\sigma^3$ and its global equality cases remain unknown here.

Computational and AI assistance

OpenAI Codex assisted with literature screening, proof search, generation and checking of the exact verifier, adversarial audit organization, and manuscript preparation. Language-model output is not treated as mathematical evidence: the finite claim rests on the displayed reduction and the released exact integer certificate and verifier. No proof assistant was used. The author is responsible for the statement and conclusions.

References

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